

Incentives & compensation

Week 39 — why paying a manager more equity makes him more cautious

Advanced Corporate Finance (EBM232A05)

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Three rows, and every one of them treated “the firm” as a single decision-maker:

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)
38	Costly external finance	investment $\downarrow$	cash hoarding (Almeida et al., 2004)
39	The manager wedge	? today	—

Today we split it in two. The person who makes the decision is not the person who owns the risk — and the gap between them has a **sign** you will not expect.

- The half you already computed: with debt outstanding, **shareholders love risk**.
- The puzzle: pay-performance sensitivity is astonishingly **weak** — and the famous correction to that number does not dissolve it.
- The twist: a risk-averse, undiversified manager takes *less* risk the **more** equity you give him.
- The fix: **convex** pay buys the risk appetite back — and can overshoot.
- Evidence: vega moves real policy, and a natural experiment that makes it **causal**.

One friction — **the manager is a person** — and it flips a sign you thought you had settled.

The half you already computed

You have done the shareholder's half

From the bachelor: once risky debt is outstanding, shareholders hold a **call** on the firm. Raise the variance of the payoff and the call is worth more — so shareholders want **more** risk, even at the cost of value. That is **asset substitution** (Jensen and Meckling, 1976), and you computed it.

- Note what that argument assumes: the person *choosing* the risk is the shareholder.
- In a public firm they are not. A manager chooses. And a manager is **one person** with **one job**, holding an undiversified claim on **one** firm.

Same conflict, different chooser — and the sign flips.

The puzzle: the incentives are weak

The weak-incentives puzzle

Jensen and Murphy (1990) ask the obvious question: when shareholder wealth rises by \$1,000, how much richer does the CEO get?

CEO wealth $\uparrow$ \$3.25 per \$1,000 of shareholder wealth.

- That is a pay-performance sensitivity of roughly **a third of one percent**. The CEO keeps essentially none of what he creates — or destroys.
- Their reading: CEOs are paid “like bureaucrats” — rewarded for **size and survival**, barely for value.

If incentives are this weak, why would any board leave them there?

...or is the number wrong?

Hall and Liebman (1998) object, and the objection is a measurement one. Look at the CEO's **holdings**, not just this year's pay: most of the sensitivity runs through the stock and options he *already owns*, which revalue every day.

- Counted that way, the elasticity of CEO wealth to firm value is about **3.3** at the median — some **20–30 times** the estimates that ignore holdings.
- So CEOs are *not* paid like bureaucrats. Incentives are real, and they grew sharply with option grants through the 1980s and 1990s.

Two careful papers, one dataset-shaped disagreement. Notice the habit: **what you measure the incentive on decides the answer.**

Why the puzzle survives the correction

Grant Hall–Liebman the point. Incentives are larger than \$3.25 suggests — but they are still **nowhere near** $\beta = 1$, and no board has ever tried to get there.

- $\beta = 1$ would mean: sell the firm to the manager. Perfect alignment, by construction.
- Nobody does it. Not because boards are asleep — because it would be **expensive**.

The two papers also measure different things — a dollar sensitivity against an elasticity — which is most of why they disagree. *Note on notation:* β today is **pay-performance sensitivity**, the manager's share of the outcome. It is not market beta.

So the real question is not “why so weak?” but **what does a strong incentive cost, and who pays it?** Answer that and the level takes care of itself.

The twist: the manager wedge

One new ingredient: the manager is a person

Take the contract as **given** and linear — salary plus a share of the outcome:

$$w = \underbrace{\alpha}_{\text{salary}} + \underbrace{\beta y}_{\text{a share } \beta \text{ of the outcome } y}, \quad 0 \leq \beta \leq 1.$$

- **Shareholders** hold a diversified portfolio: firm-specific risk washes out, so they price this project at its **NPV** and nothing else.
- **The manager** does not. His human capital, reputation and equity are all in **this** firm — he is risk-averse over exactly the risk they diversify away.

The *optimal* contract is EBM008D05's lecture, not ours. A full treatment also lets α rise to pay him for the risk imposed — a second, complementary reason $\beta < 1$. We isolate the **project-refusal** channel because it is the one you can draw.

The manager's calculus

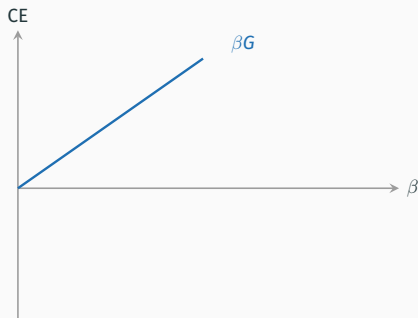
A risky project arrives: it adds $G > 0$ to expected value and $\Delta\sigma^2$ to the variance of y . Shareholders want it — $G > 0$ is the whole test. The manager evaluates his **certainty equivalent**:

$$CE(\beta) = \underbrace{\beta G}_{\text{his share of the NPV}} - \underbrace{k \beta^2}_{\text{the risk he is made to carry}}, \quad k \equiv \frac{r}{2} \Delta\sigma^2.$$

- The gain is **linear** in β ; the risk cost is **quadratic** — he holds β of the payoff, so its variance scales with β^2 . (r is his risk aversion; roll it together with $\Delta\sigma^2$ into one number k .)

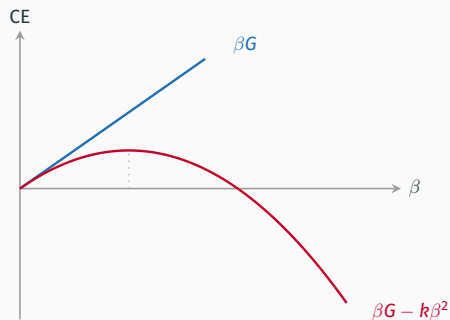
Linear against quadratic: that is the entire lecture. One wins at small β , the other at large β .

What the manager gains from a good project



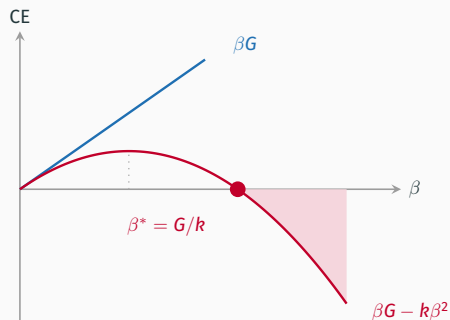
Start with the half a board would think about: give him a share β of a project worth G , and his gain rises **in proportion**. More equity, more reason to say yes. So far, alignment works.

What the manager gains from a good project



Now net off the risk he has to carry. It grows with β^2 , so it starts negligible and ends dominant: his enthusiasm **peaks**, then falls. Doubling his stake does *not* double his appetite.

What the manager gains from a good project



Past β^* it goes **negative**: he turns down a project his shareholders want, and he is right to — for him it is a bad deal. The shaded region is value destroyed **by the incentive scheme itself**.

Incentives are a cost, not a free good

Set $CE(\beta) \geq 0$ and read off the manager's investment rule:

$$\text{he takes it} \iff \underbrace{G}_{\text{its NPV}} \geq \underbrace{k\beta}_{\text{his risk cost}} \quad \text{i.e.} \quad \beta \leq \beta^* = \frac{G}{k}.$$

- The hurdle **rises in β** : stronger incentives make him **more** conservative, not more aligned. Shareholders' hurdle is $G \geq 0$, flat. The gap is the **wedge**.
- So $\beta = 1$ is not an ideal we fall short of — it is a **corner to avoid**. Weak incentives are a *choice*.

Why give him *any* equity, then? This model has only a **risk-sharing** margin; add unobservable **effort**, which $\beta > 0$ buys, and the optimum is interior. Named here, derived in EBM008D05.

Put in numbers

NPV $G = 10$, and enough added variance to give a risk-cost coefficient $k = 50$.
Shareholders want it at *every* β ; the manager's view:

Manager's stake	His share βG	His risk cost $k\beta^2$	$CE(\beta)$
$\beta = 0.05$	0.5	0.125	+0.375
$\beta = 0.10$	1.0	0.5	+ 0.5 ↑
$\beta = 0.20$	2.0	2.0	0
$\beta = 0.30$	3.0	4.5	-1.5 ↓

- Appetite **highest at 10%**, gone by 20%. Nothing about the *project* changed — only who bears its risk.
- Turn the dials: $k = 25$ (**safer**) or $G = 20$ (**better**) each double β^* to 0.4 — both buy room for stronger incentives.

Who is most exposed?

Read the comparative statics off $\beta^* = G/k$, $k = \frac{r}{2}\Delta\sigma^2$ — and they are the cross-section:

A manager who...	...and so	Risk-taking
is undiversified (wealth in the firm)	has a high r	↓
is near retirement, or holds one job	cannot re-diversify	↓
faces a genuinely volatile project ($\Delta\sigma^2 \uparrow$)	needs a bigger G	↓
was given <i>more</i> equity ($\beta \uparrow$)	carries more variance	↓
holds convex pay (options)	is paid for variance	↑

The first four all push one way, and it is the way boards do not intend. Only the last row reverses it — which is why the last row exists.

The fix, and its overshoot

Buy the risk appetite back

If the problem is that the manager dislikes variance, pay him with something whose value **rises** with variance. You know exactly such an instrument.

Stock

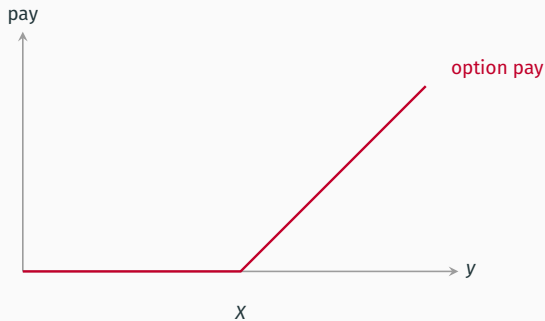
Payoff **linear** in y . Its value does not care about variance — but the *holder* does. Induces risk **aversion**.

Options

Payoff **convex** in y : the downside is cut off at the strike. Value **rises** with variance — **vega**.

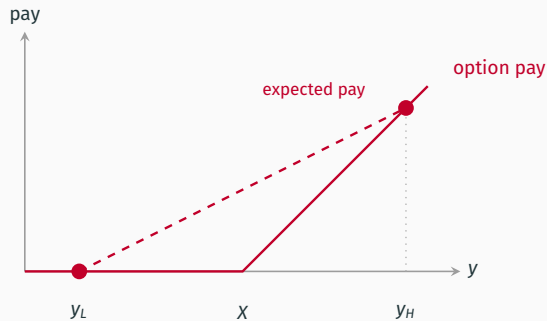
The bachelor taught you vega as a Greek to compute. Here it is a **policy instrument**: the board chooses how much risk appetite to buy. (This stock-versus-options contrast is the one thing the predecessor course said about pay; today it gets its mechanism.)

Why convexity pays for risk



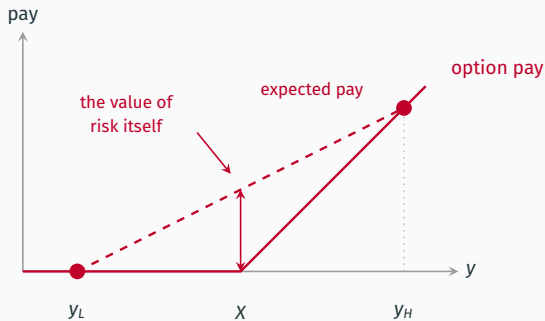
The option's payoff against the firm's outcome. Below the strike X he gets nothing; above it he gets everything. The **kink** is the instrument: losses are truncated, gains are not.

Why convexity pays for risk



Now make the outcome risky: y_L or y_H , equally likely — and note their mean is exactly the strike, so the option is **at the money**. His *expected pay* is the midpoint of the **chord** joining the two payoffs, not a point on the curve.

Why convexity pays for risk



And the chord sits **above** the curve — here the curve is flat on the axis at the strike, so the whole arrow is gain. Spreading y without touching its mean makes him strictly better off: that gap is vega. Do the same to a straight line and the gap is **zero** — which is why stock cannot do this.

...and it overshoots

Convexity does not restore the first best. It replaces one distortion with another.

- The board can dial vega until the manager accepts the projects shareholders want. Nothing stops it **overshooting**: a manager paid for variance will take risk that is **value-destroying**, because the downside is not his.
- That is asset substitution again — but now **manufactured by the pay package** rather than by leverage.
- And it lands on **creditors**: the truncation that makes options attractive is exactly the truncation they are short.

Two dials, opposite errors: too little convexity and he under-invests in risk; too much and he over-invests. No setting removes the wedge — only settings that **trade** it. Hold on to the truncation: in Week 42 the same written option belongs to the **deposit insurer**.

Evidence

Vega moves real policy

Coles et al. (2006) put the comparative static to the data: measure each CEO's **vega** (wealth sensitivity to stock *volatility*) and **delta** (sensitivity to the stock *price*), and ask what the firm then does.

Higher CEO vega is followed by...	
R&D spending	↑
investment in PP&E	↓
business-segment focus	↑
leverage	↑

Exactly the risky-policy bundle the model predicts — and they control for delta, because the two move together. They also find the reverse: riskier firms then write **higher-vega, lower-delta** contracts. Which is the warning on the next slide.

The causal test

The correlation has an obvious defence: risky firms *choose* option-heavy pay, so vega may be an effect rather than a cause. Bakke et al. (2016) break the loop with an accounting rule.

- **FAS 123R** (2006) forced firms to *expense* options. Firms that had not been expensing them cut option pay sharply — for a reason unrelated to their projects.
- Those firms then **increased their hedging** substantially, against firms that used no options or already expensed them. Take away the convexity and the manager goes back to what he wanted all along: **less risk**.

Read the design, not the coefficient: the variation comes from an **accounting standard**, not from the firm — the same move that made Chaney et al. (2012) persuasive last week.

Remark: the wedge in his head

The whole hour let the manager price the project with the **board's** numbers — the same G , the same $\Delta\sigma^2$. Nothing guarantees that he does:

- An **optimistic** manager reads G too big; a **miscalibrated** one reads $\Delta\sigma^2$ too small. Both lean *against* the wedge — part of the alignment problem dissolves in his head, for free.
- And a board that knows its manager writes the package accordingly. Turned around: the **observed contract** records what the board believes about him.

Backing that belief out of pay data — how optimistic, and how miscalibrated, boards think their CEOs are — is work in progress (Dittmann & Pfeil).

Ledger & wrap

Aligning a manager is not a matter of giving him **enough** equity. Equity transfers **risk**, and risk is exactly what an undiversified person cannot cheaply hold. Past a point, more alignment buys **less** of the behaviour you wanted.

Two things to carry forward:

- “Identify the friction” now includes **who is choosing**. The same conflict read from the manager’s chair has the **opposite sign** to the one you computed from the shareholder’s.
- The instrument that fixes it — **convexity** — is the option machinery you already own, pointed at a control problem rather than a valuation problem.

The frictions ledger — Week 39

Four rows now. Bring this back next week.

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39	The manager wedge	risk-taking ↓	weak incentives (Jensen and Murphy, 1990)

Note this row is the first whose effect runs through a **person** rather than a market. The recurring question is unchanged: **which friction, and in which direction?**

What is new here — and what is not

You already own:

- asset substitution — *shareholders* love risk
- option payoffs, convexity, vega
- diversification and firm-specific risk

This week adds:

- the **manager** wedge — and its opposite sign
- incentives as a **cost** with an interior optimum
- convexity as a **policy dial**, overshoot included

Same machinery — a payoff, a variance, a Greek — pointed at a question the bachelor never asks: what does the person choosing actually want?

Reading, and this week

- **Reading:** Jensen and Murphy (1990) — the puzzle, and still the most-argued-with number in the field. Read Hall and Liebman (1998) beside it if you want the fight.
- **Evidence to keep:** Coles et al. (2006) (vega and policy), Bakke et al. (2016) (the causal test).
- **This week:** the **mock exam** is posted — attempt it under exam conditions; we work through it in the tutorial.
- **Next:** Week 40 is the break. Week 41 brings the market for **corporate control** — what happens when the wedge grows wide enough that somebody buys the firm to close it.

Give a man a bigger share of the upside and you also hand him more of the variance — he notices.

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- Jensen, M. C. and K. J. Murphy (1990). **“Performance Pay and Top-Management Incentives”**. In: *Journal of Political Economy* 98.2, pp. 225–264.

Korteweg, A. (2010). **“The Net Benefits to Leverage”**. In: *The Journal of Finance* 65.6, pp. 2137–2170.